

Mass-Editing Memory in a Transformer, a Summary

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In the research paper, “Mass-Editing Memory in a Transformer,” the researchers introduce a new method to make mass updates to the memory of transformers used in large language models. Although methods for editing a language model’s memory existed prior to the researcher’s contribution, the existing approaches were only successful at updating up to seventy-five facts at one time. The researcher’s novel approach, Mass-Editing Memory in a Transformer, or MEMIT, outperformed the existing methods by updating up to 10,000 memories simultaneously.

Transformers are the underlying structure that powers the “memory” behind large language models. Acting as an information assembly line, transformers pass knowledge from one layer to the next, with each layer processing the information and providing a richer understanding to the next layer. Modern transformers implement a decoder-only architecture. Decoders use mathematical probabilities to predict the next word or token based on the sequence of words that came before it. For example, if the sentence, “She loves soft” is passed into a decoder, the word “pillows” should be assigned a higher probability than the word “cars” because the latent representation for the word “pillow” would be closer associated to “soft” than the latent representation for the word “cars.”

Decoders contain multiple processing layers stacked on top of each other. Information gets passed layer by layer with each layer reading the representation from the prior layer, enriching the token if necessary, and passing the token combined with the new enrichment, to the next layer. Each layer contains two main components, multi-head self-attention block and a feed forward neural network block. For the multi-head self-attention block, the token can be enriched by looking at all the previous tokens. Any of the previous tokens can provide additional context to the current token, thus enriching the token. Once the token has passed through the attention blocks it then moves to the feed forward network. The token enters the feed forward network with a rich representation of the context. Using this rich representation, the feed forward network looks for weights that are similar to the input. These weights encode patterns learned from training and it is this concept that becomes the main target in the researcher’s novel approach.

Casual Mediation Analysis (CMA) is the technique used to identify exactly which component was responsible for the source of the output. CMA is composed of three steps, a clean run, a corruption, and restoration. The clean run executes the model on the original prompt with factual information. The corruption run corrupts the input with noise. The restoration run injects the corrupted run with the activation from the clean run and analyzes

the output. If the output of that layer moves the needle towards being a closer match with the output from the clean run, that component is marked as having an effect on the memory. If the output from the layer strongly matches the output from the clean run, then the “memory” has been restored and the component or components responsible for the output have been identified.

Causal mediation analysis is what allows the researchers to identify the component responsible for source of the output. Since the researchers are able to identify the component with such a high level of precision they can also replace the value stored in that component. Part of what makes MEMIT so groundbreaking is their updating of multiple components that contribute to the “memory.” This replacement essentially allows the researchers to update the “memory” of the transformer.

In conclusion, “Mass-Editing MEMORY in a Transformer,” or MEMIT replaced the existing state of the art memory editing design. Prior to MEMIT, the maximum components updated at one time was seventy-five. With the introduction of MEMIT, researchers can now update up to 10,000 new memories, significantly advancing the field’s ability to edit memory in a transformer.